Arkaraj Mukherjee: Homework 5
!!AI WAS USED TO FIGURE OUT THE VALUES = C(...) THING AND GRIDEXTRA!!

```r
library("tidyverse");


## - Attaching core tidyverse packages ------------ tidyverse 2.0.0 -
## v dplyr     1.1.4     v readr     2.1.6
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.1     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.2
## v purrr     1.2.0
## - Conflicts ------------------------ tidyverse_conflicts() -
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts
to become errors


library("gridExtra");


##
## Attaching package:  'gridExtra'
##
## The following object is masked from 'package:dplyr':
##
##     combine


setwd("~/isi_bmath/sem_II/intro_to_statistics_and_computation_with_data");
```

1. Suppose the samples are $\mathbf{y} = (y_1, \ldots, y_n)$, for us the parameter $\theta = (\theta_1, \theta_2)$ should be in the parameter space $\mathbb{R} \times \mathbb{R}_{>0}$ with θ_1 corresponding to the mean and θ_2 the variance as usual. Then, as X_i are i.i.d. normal our likelihood function will be

$$\mathcal{L}(\mathbf{y}, \theta) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\theta_2^2}} \exp\left(-\frac{(y_i - \theta_1)^2}{2\theta_2^2}\right) = \frac{1}{(2\pi)^{n/2}\theta_2^n} \exp\left(-\frac{1}{2\theta_2^2}\sum_{i=1}^{n}(y_i - \theta_1)^2\right)$$

This is maximised in θ_1 when $\Sigma(y_i - \theta_1)^2$ is minimised as exp is increasing. Basic calculus shows that this θ_1 is achieved at $\theta_1 = \bar{\mathbf{y}}$. We can verify this is a maximum using the second derivative test on the log-likelihood with respect to θ_1: $\frac{\partial^2}{\partial\theta_1^2} \ln \mathcal{L} = -\frac{n}{\theta_2^2} < 0$, which confirms it is a maximum.

For θ_2, we want to find

$$\operatorname*{argmax}_{\theta_2 \in \mathbb{R}_{>0}} C_1\theta_2^{-n} \exp\left(-C_2\theta_2^{-2}\right) = \operatorname*{argmax}_{\theta_2 \in \mathbb{R}_{>0}} \theta_2^{-n} \exp\left(-C_2\theta_2^{-2}\right)$$

Where $C_1 > 0, C_2 \geq 0$ are the appropriate constants we can substitute in later ($C_2 = \frac{1}{2}\sum(y_i - \bar{y})^2$). We can use the derivative condition to find the θ_2 we want by setting the derivative of that w.r.t θ_2 to 0 as follows,

$$\frac{\mathrm{d}}{\mathrm{d}\theta_2}\left(\theta_2^{-n}\exp\left(-C_1\theta_2^{-2}\right)\right) = -n\theta_2^{-n-1}\exp\left(-C_2\theta_2^{-2}\right) + 2C_2\theta_2^{-n-3}\exp\left(-C_2\theta_2^{-2}\right)$$

Giving us $\theta_2^2 = (2/n)C_2 = (1/n)\Sigma(y_i - \bar{y})^2$. Thus the maximum likelihood estimate is $\hat{\theta}(\mathbf{y}) = (\bar{y}, S^2)$ where $\bar{y}$ is the sample mean and $S^2 = n^{-1}\Sigma(y_i - \bar{y})^2$ the sample variance. To confirm this is a maximum, we check the second derivative of the log-likelihood $\ell(\theta_2) = -n\ln\theta_2 - C_2\theta_2^{-2}$. We have $\ell''(\theta_2) = \frac{n}{\theta_2^2} - \frac{6C_2}{\theta_2^4}$. Substituting the critical value $\theta_2^2 = \frac{2C_2}{n}$, we get $\ell'' = \frac{n^2}{2C_2} - \frac{6C_2}{(2C_2/n)^2} = \frac{n^2}{2C_2} - \frac{3n^2}{2C_2} = -\frac{n^2}{C_2} < 0$, ensuring a maximum.

2. Suppose the samples are $\mathbf{y} = (y_1, \ldots, y_n)$, for us the parameter θ should be in the parameter space $\mathbb{R}_{>0}$ with θ corresponding to λ, then as X_i are i.i.d. poisson, the likelihood function will be

$$\mathcal{L}(\theta, \mathbf{y}) = \prod_{i=1}^{n}\frac{\theta^{y_i}\exp(-\theta)}{y_1!} = \frac{\exp(-n\theta + \ln(\theta)\sum_{i=1}^{n} y_i)}{\prod_{i=1}^{n} y_i!}$$

This is maximised when $-n\theta + \ln(\theta)\Sigma y_i = -n\theta + n\bar{\mathbf{y}}\ln(\theta)$ is maximised and we can find this maximiser like we did in the previous problem as follows,

$$\frac{\mathrm{d}}{\mathrm{d}\theta}\left(-n\theta + n\bar{\mathbf{y}}\ln(\theta)\right) = -n + \frac{n\bar{\mathbf{y}}}{\theta} = 0$$

Giving us $\theta = \bar{\mathbf{y}}$. Thus, the maximum likelihood estimate is $\hat{\theta}(\mathbf{y}) = \bar{\mathbf{y}}$. Checking the second derivative: $\frac{\mathrm{d}^2}{\mathrm{d}\theta^2}(-n\theta + n\bar{\mathbf{y}}\ln(\theta)) = -\frac{n\bar{\mathbf{y}}}{\theta^2}$. Since $\theta, \bar{\mathbf{y}} > 0$, this value is negative, confirming the estimate is a maximum.

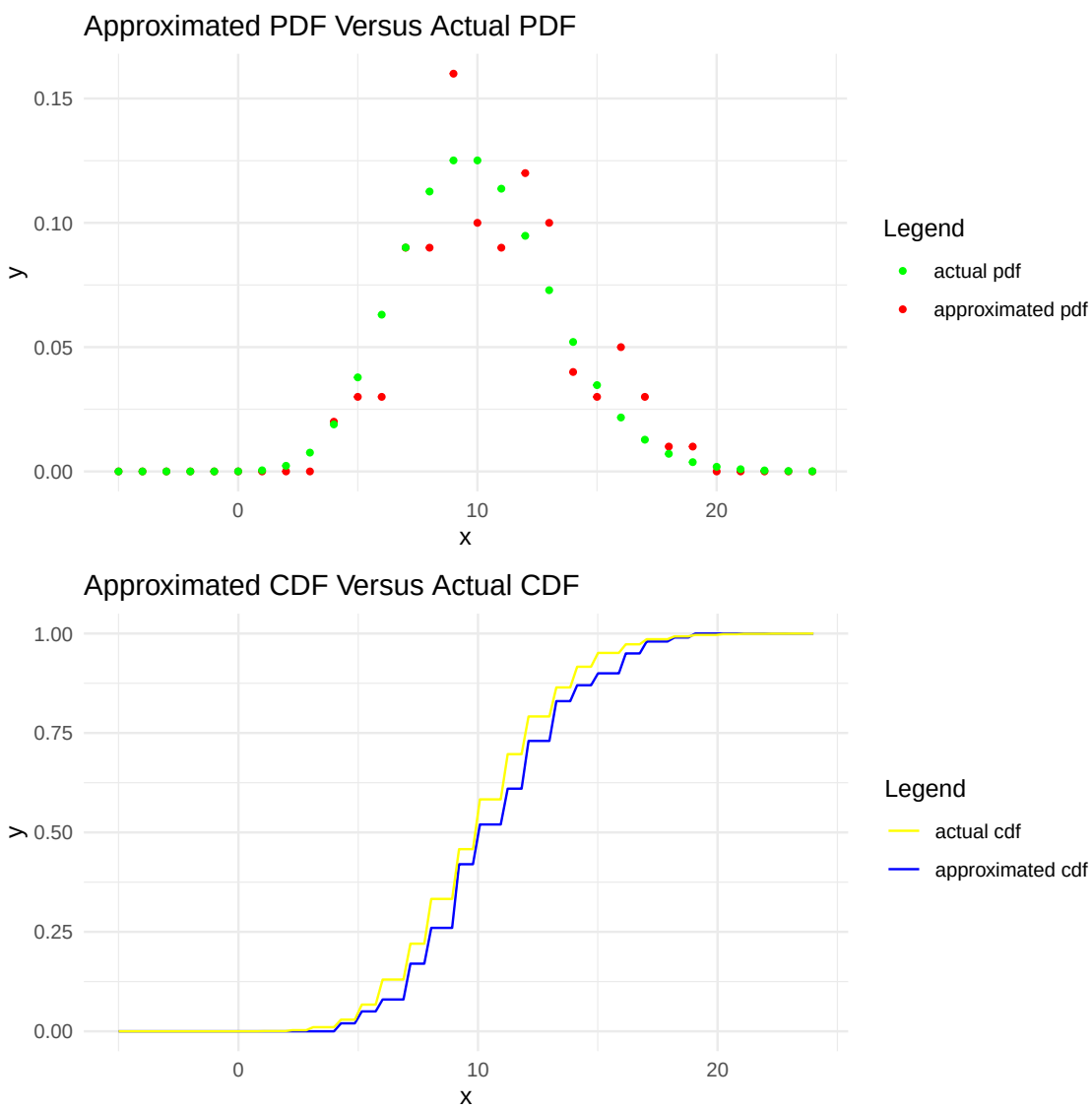
3. (a) Suppose that θ is the probability of getting heads ($\theta \in [0, 1]$) then the likelihood function here is,

$$\mathcal{L}(\theta, TTTTH) = (1-\theta)^4\theta$$

This is maximised on the boundary or where the derivative is 0 w.r.t. θ. Its seen after checking these cases that the maximum is achieved at $\theta = 1/5$. Using the second derivative test on the log-likelihood $\ell(\theta) = 4\ln(1-\theta) + \ln(\theta)$, we have $\ell''(\theta) = -\frac{4}{(1-\theta)^2} - \frac{1}{\theta^2}$. Since both terms are strictly negative, $\ell''(\theta) < 0$, confirming the maximum.

(b)
```
headest = function(N) {
    L = function(p) {(1-p)^(N-1) * p}
    optimize(L, c(0,1), maximum = TRUE)$maximum
}
# we can test this out using the previous problem
headest(5)-1/5

## [1] 1.947805e-05

# this is pretty close to zero so it works
```

4. We plot the two side by side to see how well it approximates the cdf and pdf below.

```
set.seed(13052007)
P = rpois(100, lambda = 10)
m = max(P)
ddfapx = function(v) {
    v = as.vector(v)
    unique = unique(v)
    n = length(v)
    U = as.data.frame(unique)
    fReq = Vectorize(function(val) {sum(v == val)})
    U$freq = fReq(U$unique)
    function(k) {
        if(k %in% unique) {return(U$freq[U$unique == k]/n)}
        else {return(0)}
    }
}
dcdfapx = function(v) {
    f = ddfapx(v)
    m = max(v)
    cdfvec = cumsum(Vectorize(f)(1:m))
    function(x) {
        if(floor(x) < min(v)) {return(0)}
        else if(floor(x) > m) {return(1)}
        else {return(cdfvec[floor(x)])}
    }
}
poisdfapx = ddfapx(P)
poisdcdfapx = dcdfapx(P)


to_plot = data.frame(x = -5:(m+5), y = Vectorize(poisdfapx)(-5:(m+5)))
pdf_plot = ggplot(data = to_plot, aes(x = x)) +
geom_point(aes(y = y, colour = "approximated pdf"), size = 1) +
geom_point(aes(y = dpois(x, lambda=10), colour = "actual pdf"), size = 1) +
scale_colour_manual(
  name = "Legend",
  values = c("actual pdf" = "green", "approximated pdf" = "red")
) +
ggtitle("Approximated PDF Versus Actual PDF") +
theme_minimal()

cdf_plot = ggplot(data = to_plot, aes(x = x)) +
stat_function(fun = Vectorize(poisdcdfapx),
aes(colour = "approximated cdf"), linewidth = 0.5) +
stat_function(fun = ppois, args = list(lambda=10),
aes(colour = "actual cdf"), linewidth = 0.5) +
scale_colour_manual(
```

```
  name = "Legend",
  values = c("approximated cdf" = "blue", "actual cdf" = "yellow")
) +
ggtitle("Approximated CDF Versus Actual CDF") +
theme_minimal()
grid.arrange(pdf_plot, cdf_plot, nrow=2)
```